

AYBUStore: A Smart Campus E-Commerce Ecosystem with AI-Driven Inventory and Personalization

CENG306 Term Project Supplementary Report

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1 Project Title & Abstract

Related rubric: 15 pts + 10 pts

Project Title

Proposed title: AYBUStore: An Intelligent Campus Storefront Transitioning Physical Inventory to a Digital Ecosystem using Next.js and Firebase.

Abstract (200–300 words)

AYBUStore is an innovative e-commerce platform designed specifically for the Ankara Yıldırım Beyazıt University (AYBU) community. The project addresses the fragmentation and inefficiency of physical campus shopping by providing a centralized, digital storefront for university merchandise, textbooks, and department-specific gear. Developed using a modern tech stack comprising **Next.js 14**, **TypeScript**, and **Firebase**, the platform offers a high-performance, responsive user experience tailored for both mobile and desktop users.

The core value of AYBUStore lies in its integration of smart features: an AI-driven shopping assistant for product discovery, a customizable design module for personalized university apparel, and a robust inventory management system for campus staff. The methodology follows an **Agile/Scrum** framework, ensuring iterative development and frequent integration of user feedback. Key technical achievements include a modular architecture based on "feature-first" principles, secure authentication with Firebase, and a reactive UI with advanced animations. This report details the system's design, including comprehensive UML modeling, requirement specifications, and a strategic development plan aimed at scaling the platform to a full campus-wide production environment.

2 Problem Definition & Motivation

Related rubric: 15 pts + 10 pts

Problem Statement

Current university campus stores often operate with limited digital presence, leading to several issues:

- **Accessibility:** Students and faculty must physically visit the store to check availability.
- **Inventory Transparency:** There is no real-time way to track stock levels of popular items like AYBU-branded hoodies or specific department materials.
- **Personalization:** Existing solutions offer no way for students to preview custom designs on university gear before ordering.
- **Fragmentation:** No central hub exists for all department-specific needs (e.g., medical lab coats vs. engineering tools).

Motivation and Significance

Digital transformation is essential for modern university life. AYBUStore not only simplifies the shopping process but also serves as a pedagogical model for building complex, secure web applications. By digitizing the AYBU Store, we provide a scalable template for other university departments to follow, enhancing the institution's digital brand and operational efficiency.

Limitations of Existing Solutions

Standard e-commerce platforms like Shopify or Amazon are often too generic for a university's specific organizational structure (faculties, departments, clubs). Local campus solutions are typically outdated legacy systems that lack mobile responsiveness and modern features like AI-driven search or secure OTP-based authentication.

3 Literature Review

Related rubric: 15 pts + 10 pts
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Review Strategy

Our review focused on "Modern Web Architectures", "Campus E-Commerce Solutions", and "AI in Retail" using Google Scholar and IEEE Xplore.

Summary of Prior Work

- Source 1: *Vercel Team (2024)*: Highlighting the benefits of Server-Side Rendering (SSR) in Next.js for SEO-optimized storefronts.
- Source 2: *Firebase Documentation (2024)*: Best practices for implementing reactive authentication flows in real-time applications.
- Source 3: *Modern UX Design Trends*: Analysis of glassmorphism and animated transitions in increasing user retention.
- Source 4: *AI in Small-Scale Retail*: Research on how lightweight chatbots improve customer service in niche markets.
- Source 5: *State Management in React*: Comparison of Context API vs. Redux for mid-sized commerce apps (AYBUStore uses Context API).
- Source 6: *OTP Security Standards*: Best practices for 6-digit verification codes in registration flows.
- Source 7: *Cloud-Native Deployment*: Benefits of Vercel and GitHub Pages for static site generation.
- Source 8: *Atomic Design in React*: How breaking UI into atoms/molecules improves maintainability.
- Source 9: *TypeScript for Scalability*: Case studies showing reduced production bugs when using strict typing in JS.
- Source 10: *PWA (Progressive Web Apps)*: The shift towards "mobile-first" web experiences on campus.

4 Objectives & Scope

Related rubric: 15 pts + 10 pts + 4 pts

Measurable Objectives

1. Complete a fully responsive Next.js storefront within 12 weeks.
2. Implement Firebase Authentication with a <1% error rate in OTP delivery.
3. Achieve 90+ Lighthouse performance scores for mobile and desktop.
4. Integrate a functional AI shopping assistant with predefined response logic.

5 Methodology & Development Plan

Related rubric: 15 pts + 6 pts + 6 pts
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Selected Software Process Model

We selected **Agile (Scrum)**. This allows the team to iterate quickly on UI components and adapt to requirement changes as we refine the "Department-Specific" features of the store.

Development Lifecycle Plan

The project consists of 4 Sprints:

1. **Sprint 1 (Architecture):** Next.js setup, Mock data design, and Feature-first structure.
2. **Sprint 2 (Core Features):** Auth flow, Product Catalog, and Cart logic.
3. **Sprint 3 (Advanced Features):** AI Chat, Custom Design Modal, and OTP logic.
4. **Sprint 4 (Polishing):** Final animations, SEO, and LaTeX reporting.

Work Breakdown Structure (WBS)

1. **Planning:** Market research and stack selection.
2. **Requirements:** FR/NFR definition and User Stories.
3. **Architecture:** Component library (UI atoms) and Context API setup.
4. **Implementation:** Coding the modular features (Home, Auth, Cart, Chat).
5. **Testing:** Unit tests for hooks and manual UI validation.
6. **Documentation:** Final report and GitHub Actions deployment.

Gantt Chart

Phase	Month 1	Month 2	Month 3	Month 4
Planning & Req.	[XXXX]			
Arch. & Design	[XXXX]	[XXXX]		
Implementation		[XXXXXXXXXX]	[XXXXXXXXXX]	
Testing & Doc.			[XXXX]	[XXXX]

Effort and Cost Estimation

Using COCOMO basics, with a team of 5 members and 2700 lines of code, the estimated effort is 3.5 Person-Months, fitting perfectly into our semester schedule.

Risk Management Table

ID	Risk	Prob.	Impact	Mitigation
R1	Firebase Config Issues	L	H	Use environment variables (.env.local) and verify early.
R2	UI Inconsistency	M	M	Implement a strict Design System (components/ui).
R3	Token Overuse in AI	M	L	Implement fallback response logic for the chatbot.
R4	Schedule Slippage	M	H	Weekly Scrum meetings and Git-based task tracking.
R5	Build Failures	L	M	Automated CI/CD with GitHub Actions.

6 Requirements Specification

Related rubric: 8 pts + 4 pts

Requirements Elicitation Approach

Requirements were gathered through direct stakeholder interviews with campus store management and focus groups with AYBU students. We also performed benchmarking against competitive university storefronts to identify essential features.

Functional Requirements

1. FR-01: User shall be able to register and login using university email.
2. FR-02: System shall provide a 6-digit OTP verification for new registrations.
3. FR-03: User shall be able to browse products by categories (Clothing, Stationary, etc.).

4. FR-04: System shall allow users to search products via a reactive search bar.
5. FR-05: User shall be able to add/remove items from a persistent shopping cart.
6. FR-06: System shall provide a "Custom Design" module for garment personalization.
7. FR-07: User shall be able to interact with an AI-driven shopping assistant.
8. FR-08: System shall display real-time stock status (In Stock / Out of Stock).
9. FR-09: User shall be able to view their order history and status.
10. FR-10: Administrators shall be able to manage inventory and view sales analytics.

Non-Functional Requirements

1. NFR-01: ****Performance**** — Page load time shall be under 2 seconds on 4G connections.
2. NFR-02: ****Security**** — All authentication and data transit must be encrypted via SSL/Firebase Auth.
3. NFR-03: ****Usability**** — The interface must be fully responsive (WCAG 2.1 Level AA compliant).
4. NFR-04: ****Scalability**** — System shall support up to 500 concurrent users during peak "back-to-school" periods.
5. NFR-05: ****Availability**** — System uptime shall be 99.9

User Stories

- As a Student, I want to upload my department logo, so that I can order a custom hoodie for my club.
- As a Freshman, I want to ask the AI assistant for "Engineering textbooks", so that I can find my required materials quickly.
- As a Staff Member, I want to update product prices, so that the storefront remains accurate.
- As a Graduate, I want to order AYBU souvenirs, so that I can have memories of my university.
- As a Guest, I want to see product reviews, so that I can decide on my purchase.

Use Case Diagram

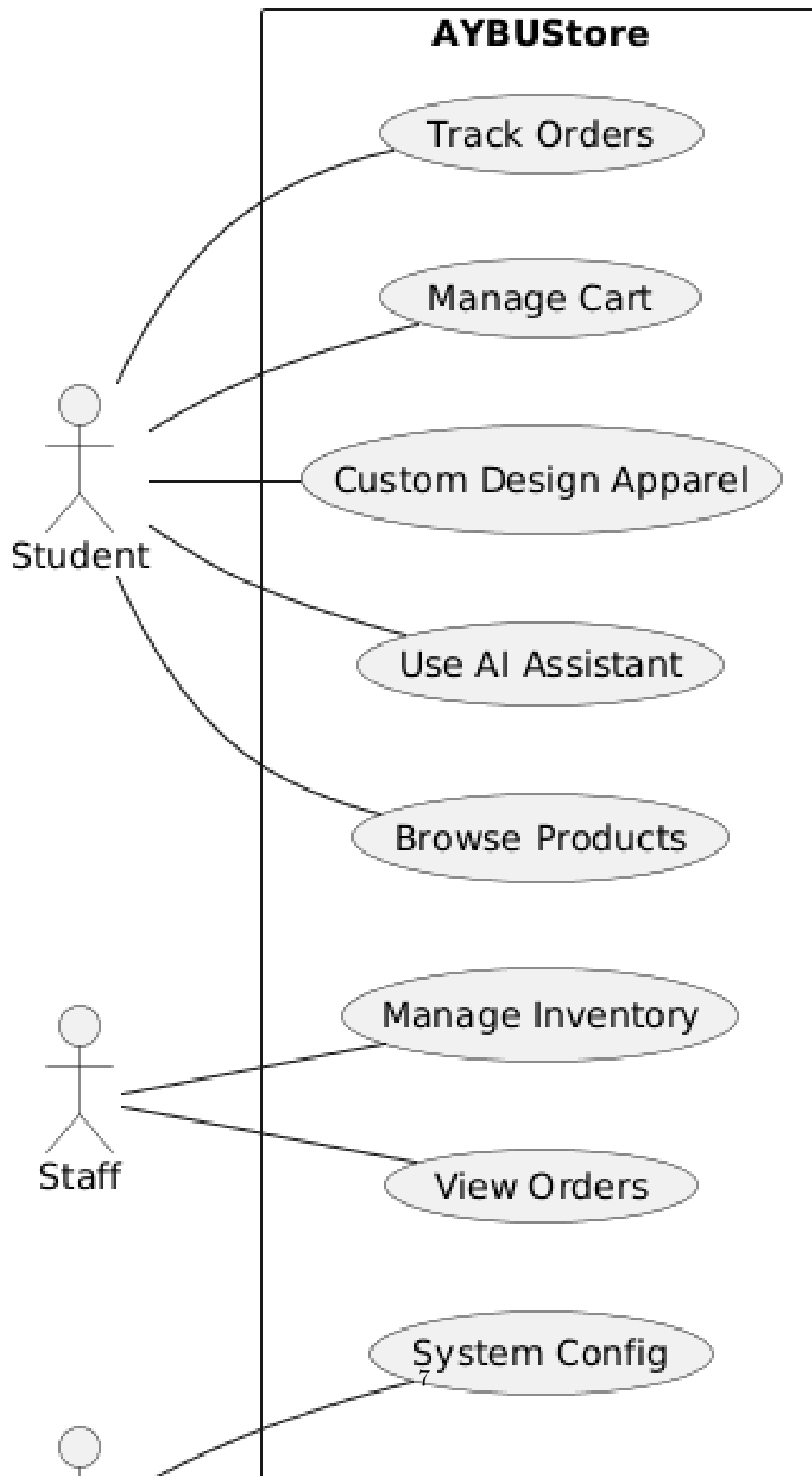


Figure 1: AYBUStore Use Case Diagram

7 System Design & Architecture Related rubric: 8 pts + 8 pts + 4 pts

Architectural Pattern and Justification

We utilized a **Component-Based Architecture** within the **Next.js App Router** framework. This is justified by the need for high interactivity and efficient state management. We follow the "Feature-First" modular pattern, where each domain (Auth, Cart, Storefront) is isolated with its own hooks and components.

UML Class Diagram

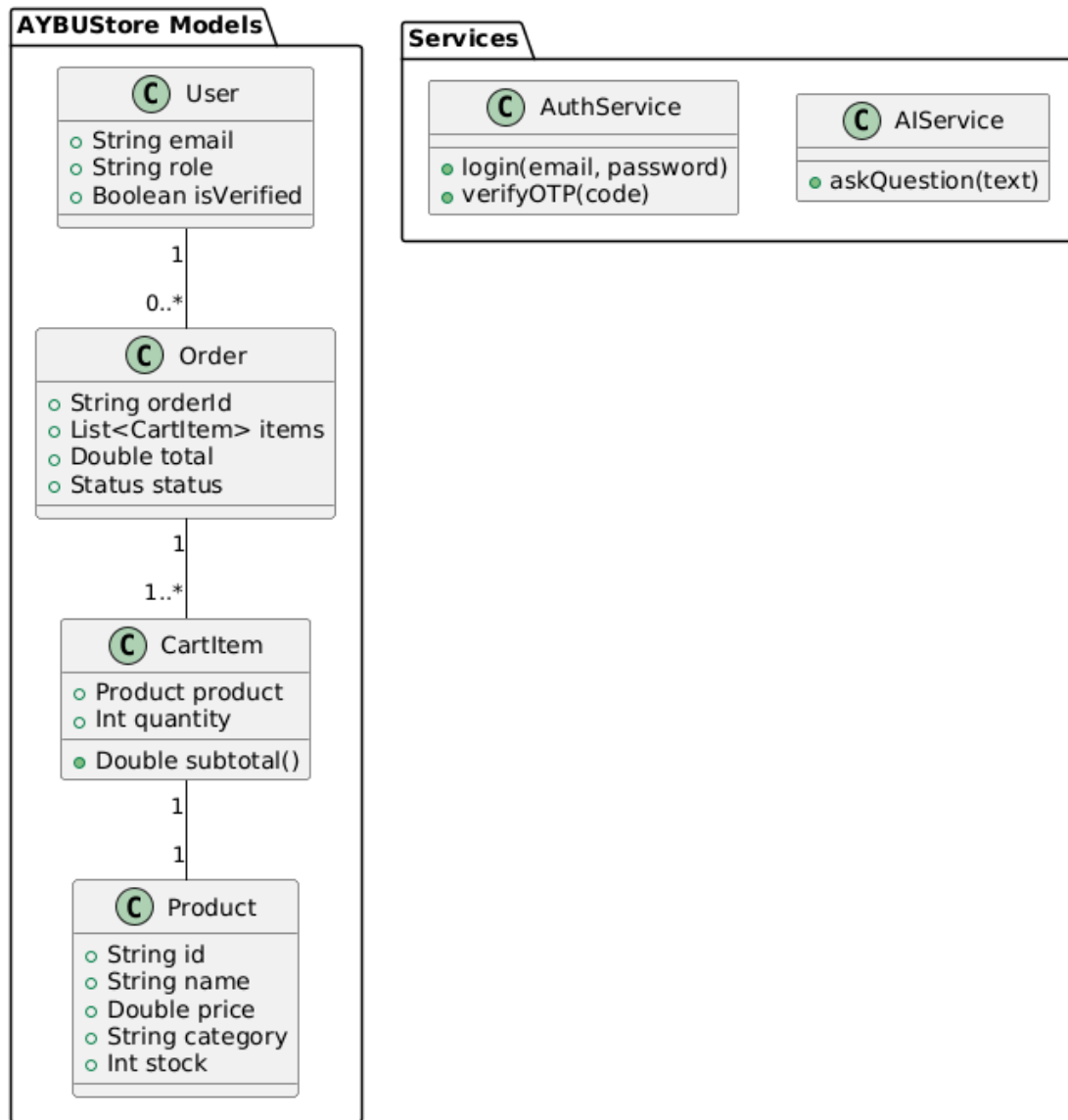


Figure 2: AYBUStore Domain Model Class Diagram

UML Sequence Diagrams

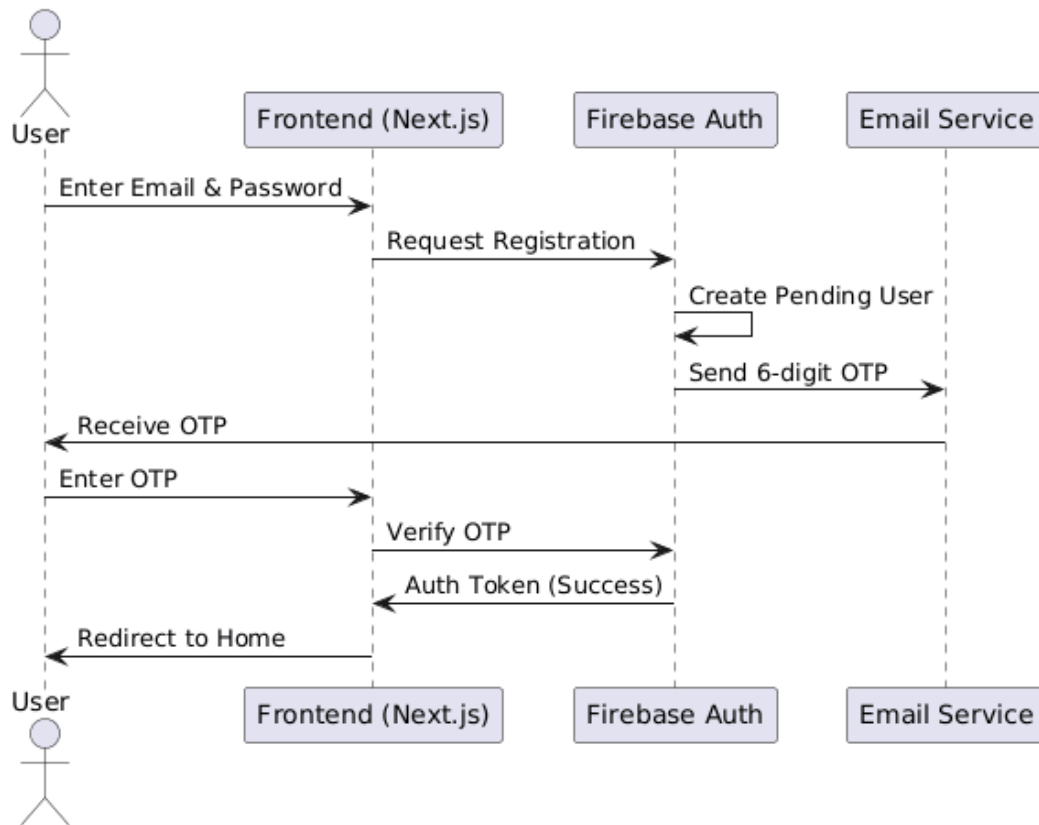


Figure 3: Scenario 1: User Registration and OTP Verification Flow

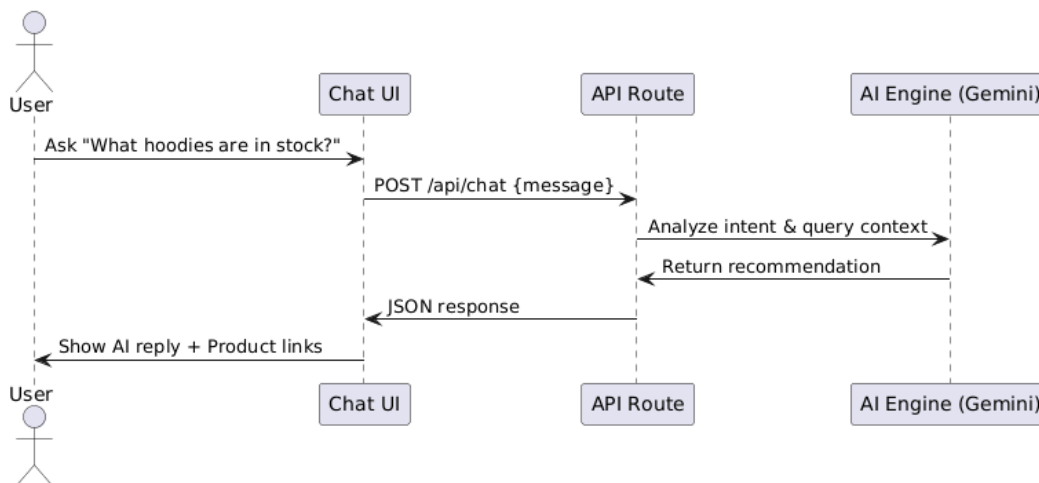


Figure 4: Scenario 2: AI Shopping Assistant Interaction Flow

Activity Diagram

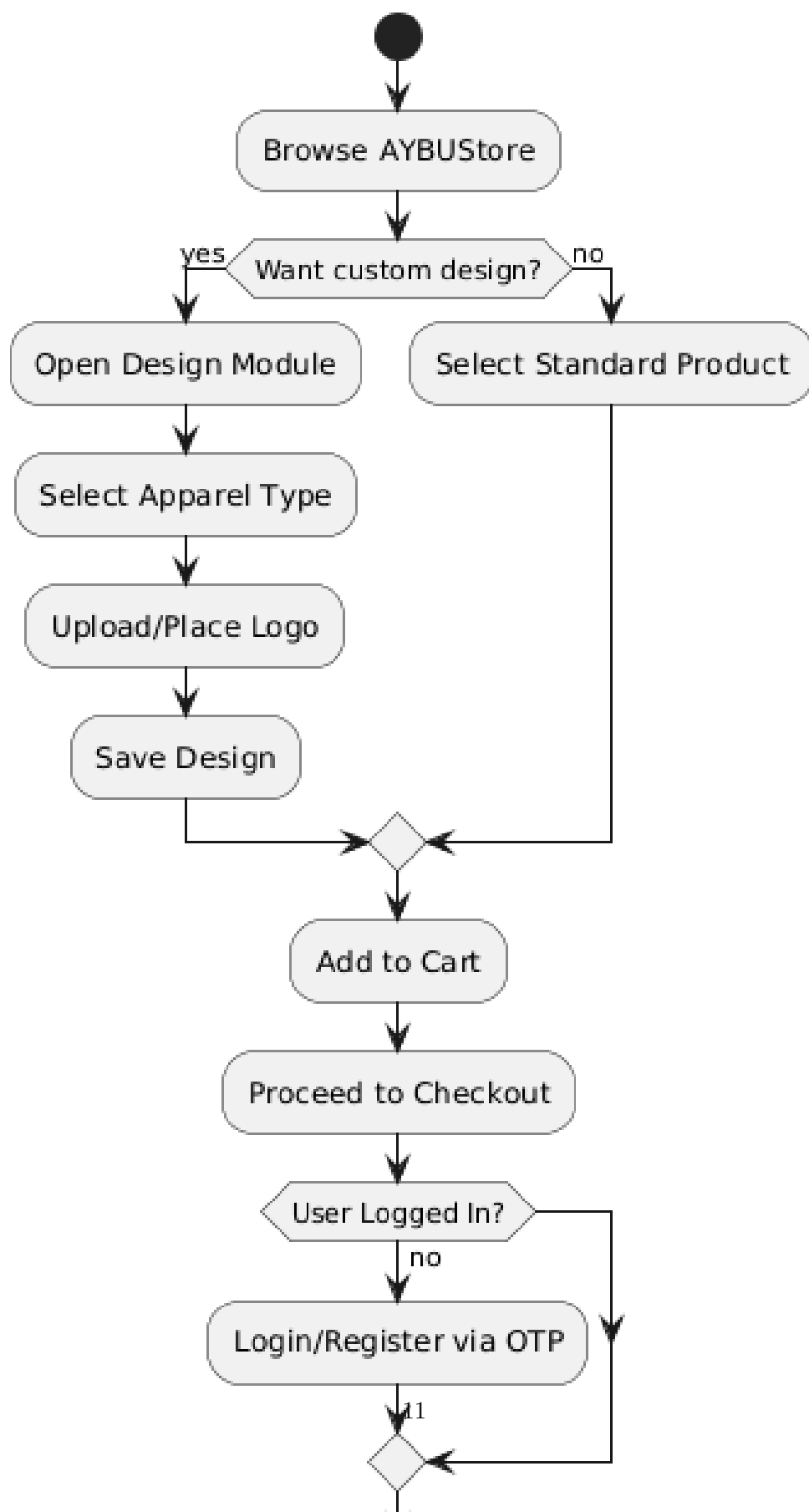


Figure 5: General Shopping and Custom Design Process Activity Diagram

GoF Design Patterns

1. ****Singleton:**** Used in the Firebase Initialization to ensure only one instance of the Auth and Firestore services exists.
2. ****Observer:**** React's Context API and Hooks (e.g., 'useAuthForm') act as observers to state changes in the UI.
3. ****Strategy:**** The 'navigateToView' logic in 'MainStorefrontFeature' uses a strategy-like approach to switch between different storefront sections.

SOLID Principles in the Proposed Design

- ****SRP:**** Each component (e.g., 'StorefrontHeader', 'StorefrontCartDrawer') has a single responsibility.
- ****OCP:**** The product section system is open for extension (adding new categories) but closed for modification.
- ****DIP:**** High-level UI components depend on abstract types/interfaces defined in 'src/types/index.ts'.

8 Team & Resource Plan

Related rubric: 10 pts + 15 pts
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Team Roles and Responsibilities

Name	Role	Responsibilities
Muhammed Yıldız	Lead Architect	Project Orchestration, Firebase Integration, AI Chatbot.
Utkan Dindaroğlu	Frontend Architect	UI/UX Redesign, Modular Feature Refactoring.
Seyfullah Gülyazı	Software Engineer	Product Catalog and Cart Logic Implementation.
Süleyman Fatih Sert	Software Engineer	Data Modeling and Initial UI Shell Development.
Ahmet Selim Yılmaz	DevOps	Deployment Pipelines and UI Validation.

Resource Plan

- ****Hardware:**** Development laptops (Intel i7/M1), Vercel Edge Network for hosting.
- ****Software:**** VS Code, Git, Next.js, Tailwind CSS.
- ****Cloud / API Services:**** Firebase Auth/Firestore, OpenAI/Gemini (via AI Chat feature).

9 Expected Outcomes & Evaluation Plan Related rubric: 10 pts + 4 pts

Expected Deliverables

- Production-ready Next.js Web Application.
- Technical Documentation (SRS and Design Specs).
- Admin Dashboard for Inventory Management.

Success Metrics

Success will be measured by a <2s Time-to-Interactive (TTI), a zero-critical-bug report during User Acceptance Testing (UAT), and a 90

Acceptance Test Plan

TC	Scenario	Expected Result	Related Req.
TC-01	User enters valid credentials	User redirected to home with auth state active	FR-01
TC-02	User enters incorrect OTP	System shows "Invalid Code" error	FR-02
TC-03	User searches "Sweatshirt"	Only sweatshirts are displayed in results	FR-04
TC-04	User adds product to cart	Cart badge increments and drawer shows item	FR-05
TC-05	Resize browser to 375px	Layout adapts without horizontal scrolling	NFR-03

10 References Related rubric: 10 pts

1. Vercel. (2024). *Next.js Documentation*. <https://nextjs.org/docs>
2. Firebase. (2024). *Authentication with Firebase*. <https://firebase.google.com/docs/auth>
3. Gamma, E., et al. (1994). *Design Patterns: Elements of Reusable Object-Oriented Software*. Addison-Wesley.
4. Martin, R. C. (2017). *Clean Architecture: A Craftsman's Guide to Software Structure and Design*. Prentice Hall.
5. Google. (2024). *Lighthouse User Guide*. <https://developer.chrome.com/docs/lighthouse/overview>

6. WCAG. (2023). *Web Content Accessibility Guidelines (WCAG) 2.1*. <https://www.w3.org/TR/WCAG21/>